

The ProbLemma Channel Integrals

This document links all the integrals evaluated on the ProbLemma channel to the episodes in which these evaluations were carried out.



In addition, all the integral related episodes are collected in the "Integrals Are Us" play list.

1. Season 2 Episode 15 (S2M17): Fourier Series of $\log(\Gamma(x))$ over $(0,1]$

$$\int_0^1 \log(\Gamma(x))\,dx = \log(\sqrt{2\pi})$$

$$\int_0^1 \log(\Gamma(x))\cos(2n\pi x)\,dx = \frac{1}{4n},\ n = 1,2,3,\ldots$$

$$\int_0^1 \log(\Gamma(x))\sin(2n\pi x)\,dx = \frac{\gamma + \log(2n\pi)}{2n\pi},\ n = 1,2,3,\ldots$$

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2. Season 2 Episode 19 (S2M21): Effectiveness Of Advertisement (Equation)

$$\int \frac{1}{x(a-x)}\,dx = \frac{1}{a}\log\left(\left|\frac{x}{a-x}\right|\right) + C$$

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3. Season 2 Episode 20 (S2M22): Fresnel Integrals Via Equations (Equation)

$$\int_0^{+\infty} e^{-ax^2}\cos(bx^2)\,dx = \frac{1}{2}\sqrt{\frac{\pi}{2}}\cdot\sqrt{\frac{a+\sqrt{a^2+b^2}}{a^2+b^2}},\ a>0, b\geqslant 0$$

$$\int_0^{+\infty} e^{-ax^2}\sin(bx^2)\,dx = \frac{1}{2}\sqrt{\frac{\pi}{2}}\cdot\sqrt{\frac{-a+\sqrt{a^2+b^2}}{a^2+b^2}},\ a>0, b\geqslant 0$$

$$\int_0^{+\infty} \cos\left(x^2\right) dx = \int_0^{+\infty} \sin\left(x^2\right) dx = \frac{1}{2}\sqrt{\frac{\pi}{2}}$$

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4. Season 2 Episode 24 (S2M26): Integral Evaluation via Scope Reduction (Scope Reduction)

$$\int_c^{2c} \frac{x}{\sqrt{x^2+cx-2c^2}}\, dx = 2c - \frac{c}{2}\log(3)$$

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5. Season 2 Episode 30 (S2M32): One Integral? Two Integrals! (Scope Expansion)

$$\int e^{ax}\sin(bx)\cos(cx)\, dx =$$

$$= e^{px}\left(\frac{a\sin((b+c)x) - (b+c)\cos((b+c)x)}{a^2+(b+c)^2} + \frac{a\sin((b-c)x) - (b-c)\cos((b-c)x)}{a^2+(b-c)^2}\right) + C$$

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6. Season 2 Episode 43 (S2M45): A Poisson Integral By The Book (Divide And Conquer)

$$P_{|r|\leqslant 1} = \int_0^{\pi} \log\left(1 - 2r\cos(x) + r^2\right) dx = 0, \quad |r| \leqslant 1$$

$$P_{|r|>1} = \int_0^{\pi} \log\left(1 - 2r\cos(x) + r^2\right) dx = 2\pi\log(|r|), \quad |r| > 1$$

$$\int_0^{\pi} \log(2 - 2\cos(x))\, dx = 0$$

$$\int_0^{\pi} \log(2 + 2\cos(x))\, dx = 0$$

The Euler's "log-sin" integral is evaluated in this episode not "by the book", via a limit of an integral sum, but via Euler's clever and witty substitutions.

The evaluation of the Euler's integral "by the book" is done in a different episode, see below:

$$\int_0^{\frac{\pi}{2}} \log\left(\sin\left(t\right)\right) dt = -\frac{\pi}{2} \log\left(2\right)$$

$$\int_0^{\frac{\pi}{2}} \log\left(\cos\left(t\right)\right) dt = -\frac{\pi}{2} \log\left(2\right)$$

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7. Season 2 Episode 53 (S2M53): Lanchester's Square Law (Mathematical Equivalence)

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left(\frac{x}{a} + \sqrt{\frac{x^2}{a^2}+1}\right) + C$$

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8. Season 2 Episode 54 (S2P2): Slitating Or Roliding Bead (Mathematical Equivalence)

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left(\frac{x}{a} \pm \sqrt{\frac{x^2}{a^2}-1}\right)$$

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9. Season 3 Episode 2 (S3M1): An Integral Transformation

For any whole positive number n and any real numbers $x > 1$ and θ it is the case that:

$$\int_0^{\pi} \left(x + \sqrt{x^2-1}\cos\left(\theta\right)\right)^n \, \mathrm{d}\theta = \int_0^{\pi} \frac{\mathrm{d}\theta}{\left(x - \sqrt{x^2-1}\cos\left(\theta\right)\right)^{n+1}}$$

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10. Season 3 Episode 12 (S3M7): Integration Of Binomial Differentials

If:

$$J(p,q) = \int (a+bz)^p z^q \, dz, \quad p,q \in \mathbb{Q}$$

then, for $p \neq -1$:

$$J(p,q) = -\frac{(a+bz)^{p+1} z^{q+1}}{a(p+1)} + \frac{p+q+2}{a(p+1)} J(p+1,q)$$

for $q \neq -1$:

$$J(p,q) = \frac{(a+bz)^{p+1} z^{q+1}}{a(q+1)} - b\frac{p+q+2}{a(q+1)} J(p,q+1)$$

for $p+q \neq -1$:

$$J(p,q) = \frac{(a+bz)^{p} z^{q+1}}{p+q+1} + \frac{ap}{p+q+1} J(p-1,q)$$

and:

$$J(p,q) = \frac{(a+bz)^{p+1} z^{q}}{b(p+q+1)} - \frac{aq}{b(p+q+1)} J(p,q-1)$$

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11. Season 3 Episode 24 (S3M13): 3 Creative Evaluations Of Ubiquitous Integrals

$$\int e^{ax} \cos(bx)\, dx = e^{ax} \frac{a\cos(bx) + b\sin(bx)}{a^2+b^2} + C$$

$$\int e^{ax} \sin(bx)\, dx = e^{ax} \frac{a\sin(bx) - a\cos(bx)}{a^2+b^2} + C$$

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12. Season 3 Episode 30 (S3P7): A Loxodrome, A Deduction Of An Equation

$$\int \frac{dx}{\sin(x)} = \log\left|\tan\left(\frac{x}{2}\right)\right| + C$$

$$\int \frac{dx}{\cos(x)} = \int \frac{d\left(x+\frac{\pi}{2}\right)}{\sin\left(x+\frac{\pi}{2}\right)} = \log\left|\tan\left(\frac{x}{2}+\frac{\pi}{4}\right)\right| + C$$

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13. Season 3 Episode 32 (S3M17): Hyperbolically Speaking

$$\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right| + C$$

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14. Season 3 Episode 33 (S3M18): Recurrence Relations Evaluate Integrals

$$\int_0^{\frac{\pi}{2}} \sin^m(x)\,dx = \int_0^{\frac{\pi}{2}} \cos^m(x)\,dx = \frac{(m-1)!!}{m!!}\cdot\frac{\pi}{2}, \quad m = 2n$$

$$\int_0^{\frac{\pi}{2}} \sin^m(x)\,dx = \int_0^{\frac{\pi}{2}} \cos^m(x)\,dx = \frac{(m-1)!!}{m!!}, \quad m = 2n+1$$

$$\int_0^1 \arcsin\left(x^{\frac{1}{m}}\right)dx = \int_0^1 \arccos\left(x^{\frac{1}{m}}\right)dx = \frac{\pi}{2} - \frac{(m-1)!!}{m!!}\cdot f$$

where:

$$\mathrm{f} = \begin{cases} \frac{\pi}{2} & \text{if } m \text{ is even: } m = 2n \\ 1 & \text{if } m \text{ is odd: } m = 2n+1 \end{cases}$$

$$\int_0^a \left(a^2-x^2\right)^n dx = a^{2n+1}\frac{(2n)!!}{(2n+1)!!}$$

$$\int_0^a \left(a^2-x^2\right)^{\frac{2n-1}{2}} dx = a^{2n}\frac{(2n-1)!!}{(2n)!!}\cdot\frac{\pi}{2}$$

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15. Season 3 Episode 38 (S3M21): What Mathematicians Do All Day

$$\int_0^{+\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$$

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16. Season 3 Episode 54: Partial Fraction Decompositions Evaluate Integrals

$$\int_0^{+\infty} \frac{\sin(x)}{x}\,dx = \frac{\pi}{2}$$

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17. Season 3 Episode 55 (S3P13): Kinematics In A Barn Or A Ladybug On A Rake

$$\int \sqrt{r^2 - x^2}\,dx = \frac{x}{2} \cdot \sqrt{r^2 - x^2} + \frac{r^2}{2} \cdot \arcsin\left(\frac{x}{r}\right) + C$$

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18. Season 3 Episode 57 (S3M30): Frullani's Integrals Forever

Theorem 1. If a single-variable real-to-real function $f(x)$ is 1) defined and continuous for all $x \geqslant 0$ and 2) the following *finite* limit exist:

$$\lim_{x \to +\infty} f(x) = f(+\infty) \tag{1}$$

then for all real numbers $a, b > 0$ it is the case that:

$$\int_0^{+\infty} \frac{f(ax) - f(bx)}{x}\,dx = [f(0) - f(+\infty)] \cdot \log\left(\frac{b}{a}\right) \tag{2}$$

where the expression $f(+\infty)$ is understood precisely in the sense of the limit in (**1**) and $\log()$ is the logarithm in base e.

Theorem 2. If a single-variable real-to-real function $f(x)$ is 1) defined and continuous for all $x \geqslant 0$ and 2) does not posses a finite limit (**1**) but for a finite positive real number $c > 0$ there exists an integral:

$$\int_{c}^{+\infty} \frac{f(x)}{x}\, dx \tag{3}$$

then for all real numbers $a, b > 0$ it is the case that:

$$\int_{0}^{+\infty} \frac{f(ax) - f(bx)}{x}\, dx = f(0) \cdot \log\left(\frac{b}{a}\right) \tag{4}$$

where $\log()$ is, again, the logarithm in base e.

Theorem 3. If the above function $f(x)$ is not continuous at $x = 0$ but for some finite positive real number c there exists an integral:

$$\int_{0}^{c} \frac{f(x)}{x}\, dx \tag{5}$$

then for all real numbers $a, b > 0$ it is the case that:

$$\int_{0}^{+\infty} \frac{f(ax) - f(bx)}{x}\, dx = f(+\infty) \cdot \log\left(\frac{a}{b}\right) \tag{6}$$

where the expression $f(+\infty)$ is understood precisely in the sense of the limit in (**1**) and $\log()$ is the logarithm in base e.

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19. Season 3 Episode 58 (S3M31): Improper Integrals Of Rational Functions Or The Real Granddaddy Of CRT

An improper integral over the entire real line of rational function that has the following shape:

$$\frac{P_n(x)}{Q_m(x)}$$

where:

- the degree n of the polynomial $P_n(x)$ is by at least two units smaller than the degree m of the polynomial $Q_m(x)$, $n + 2 \leqslant m$, and
- the polynomial $Q_m(x)$ has no real roots

is given by:

$$\int_{-\infty}^{+\infty} \frac{P_m(x)}{Q_n(x)}\, dx = 2\pi \cdot i \cdot \sum_k A_k^+$$

where the constant numbers A_k^+ are the coefficients of the partial fractions decomposition of the given integrand over the complex roots of the polynomial $Q_n(x)$ that correspond to these distinct roots that have strictly positive imaginary parts. Geometrically these roots correspond to the points in the upper half of the complex plane, the points that live above the real axis.

As a sample application of this technology, we showed that:

$$\int_{-\infty}^{+\infty} \frac{1}{1+x^4}\, dx = \frac{\pi\sqrt{2}}{2}$$

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20. Season 3 Episode 65 (S3M34): Integrate By Differentiation

$$\int \frac{x^2}{(x\sin(x)+\cos(x))^2}\, dx = \frac{\sin(x) - x\cos(x)}{x\sin(x)+\cos(x)} + C$$

Applying pleasant integration boundaries to the above result, we showed that:

$$\int_0^{\pi/2} \frac{x^2}{(x\sin(x)+\cos(x))^2}\, dx = \frac{2}{\pi}$$

$$\int_0^{\pi/3} \frac{x^2}{(x\sin(x)+\cos(x))^2}\, dx = \frac{3\sqrt{3}-\pi}{3+\pi\sqrt{3}}$$

$$\int_0^{\pi/4} \frac{x^2}{(x\sin(x)+\cos(x))^2}\, dx = \frac{4-\pi}{4+\pi}$$

and that:

$$\int_0^{\pi/6} \frac{x^2}{(x\sin(x)+\cos(x))^2}\,dx = \frac{6-\pi\sqrt{3}}{6\sqrt{3}+\pi}$$

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21. Season X Episode YY (SxMyy): Lorem Ipsum